Module 06 Lab Exercise - Contribution Journal

ITAI 1371 - Introduction to Machine Learning

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I worked with the Titanic dataset, prepared the data, defined the features and target variables, and split the dataset into training and testing portions. I then created, trained, and evaluated a linear regression model to predict passenger ticket fares.

For the classification portion, I converted the sex column into numeric values, selected age, passenger class, and sex as predictors, and used survival as the target. I created and trained a logistic regression model, generated predictions, and evaluated its performance using accuracy_score. The model returned a survival prediction accuracy of 74.83%.

I also completed the Knowledge Check questions. I explained the difference between regression and classification, ran code to display the linear regression coefficients, interpreted what those coefficients meant, and explained why mean squared error and accuracy are appropriate for different prediction tasks.

Throughout the lab, I checked my code, corrected spelling errors that caused execution failures,
reran the cells, and verified the results. I also adjusted the Markdown formatting so my written responses displayed correctly. My contribution included coding, troubleshooting, interpreting the model results, answering the questions, and saving the completed lab as a PDF for submission. This work strengthened my understanding of the complete process from data preparation through model evaluation.

John Grubbs

I independently completed and ran my copy of the Module 6 Regression and Classification Models lab using the Titanic dataset. For the regression portion, I removed rows with missing Age values and used Age and Pclass as features to predict passenger Fare. I split the data into training and test sets, created a LinearRegression model, trained it, and generated predictions on the test data.

I evaluated the linear-regression model with mean squared error and obtained an MSE of approximately 3364.92. I also reviewed the square root of the MSE, about $58.01, to better understand the typical size of the prediction error in dollars. Finally, I printed and interpreted the model coefficients to see how changes in Age and passenger class related to the predicted fare.

For the classification portion, I converted the Sex column into numerical values and used Age, Pclass, and Sex to predict whether a passenger survived. I trained a LogisticRegression model and evaluated it with accuracy_score, achieving approximately 74.83% accuracy. I also completed the knowledge check in my own words, explaining the differences between regression and classification, model coefficients, and the appropriate evaluation metrics for each task.